

SKILL WEEK 7

Task 1:

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {

    pid_t pid;
    int status;

    pid = fork();
    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {

        // Child process
        printf("Child process started.\n");
        printf("Child PID: %d\n", getpid());

        sleep(3);

        printf("Child process completed.\n");

        exit(0);

    } else {

        // Parent process
        printf("Parent process waiting for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
            printf("Child exited normally.\n");
            printf("Child exit status: %d\n", WEXITSTATUS(status));
        }
    }
}
```

```

        printf("Parent process completed.\n");
    }

    return 0;
}

```

```

venkat038@AYYAPPA:~$ nano 7.1.c
venkat038@AYYAPPA:~$ nano s7.1.c
venkat038@AYYAPPA:~$ nano s7.1.c
venkat038@AYYAPPA:~$ gcc s7.1.c -o s7.1
venkat038@AYYAPPA:~$ ./s7.1
Parent process waiting for child...
Child process started.
Child PID: 1243
Child process completed.
Child exited normally.
Child exit status: 0
Parent process completed.
venkat038@AYYAPPA:~$ |

```

Task 2:

Code:

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

int main() {

    char *path = getenv("PATH");

    if (path == NULL) {        printf("PATH
variable not found.\n");    return 1;
    }

    printf("PATH:\n%s\n\n", path);

    char *copy = strdup(path);
    char *directory = strtok(copy, ":");

    char filename[100];

    printf("Enter executable name to search: ");
    scanf("%99s", filename);
}

```

```

while (directory != NULL) {

    char fullpath[500];

    snprintf(fullpath, sizeof(fullpath),
             "%s/%s", directory, filename);

    if (access(fullpath, X_OK) == 0) {
        printf("Executable found: %s\n", fullpath);

free(copy);
return 0;
    }

    directory = strtok(NULL, ".");
}

printf("Executable not found in PATH.\n");

free(copy);

return 0;
}

```

Output:

```

venkat038@AYYAPPA:~$ nano s7.2.c
venkat038@AYYAPPA:~$ gcc s7.2.c -o s7.2
venkat038@AYYAPPA:~$ ./s7.2
PATH:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Users/Administrator/AppData/Local/Microsoft/WindowsApps:/mnt/c/Program Files/HP/OMEN-Broadcast/Common:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Server 8.0/bin:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/venka/AppData/Local/Programs/Python/Python310/Scripts:/mnt/c/Users/venka/AppData/Local/Programs/Python/Python310:/mnt/c/Users/venka/AppData/Local/Microsoft/WindowsApps:/mnt/c/CPROG/S10C/bin:/mnt/c/Users/venka/Downloads/jdk-25_windows-x64_bin.zip/jdk-25/bin:/mnt/c/Program Files/Java/jdk-25/bin:/mnt/c/Users/venka/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/venka/AppData/Roaming/npm:/mnt/c/Users/venka/AppData/Local/Kingsoft/WPS Office/12.1.0.27458/clitool:/snap/bin

Enter executable name to search: ls
Executable found: /usr/bin/ls
venkat038@AYYAPPA:~$

```